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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

TECHNICAL PRESENTATION

STEGANOGRAPHY VS CRYPTOGRAPHY: HIDDEN SECURITY IN THE DIGITAL WORLD

CSA5112 – CRYPTOGRAPHY AND NETWORK SECURITY

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INTRODUCTION

- Cryptography protects information by making its content unreadable without the correct key.
- Steganography hides a message inside an ordinary-looking image, audio, video, or text file.
- Plaintext is the original readable message; ciphertext is the encrypted message.
- Cryptography hides the meaning of a message, while steganography hides the existence of a message.
- Both techniques can be combined to add another layer of digital security.



CRYPTOGRAPHY

- Cryptography transforms plaintext into ciphertext using an encryption algorithm and a key.
- Only someone with the correct key should be able to recover the original message.
- Encryption protects the meaning of information while it is stored or transmitted.
- Modern examples include AES for data encryption and TLS for secure web communication.
- Security depends on strong algorithms, safe key management, and correct implementation.



STEGANOGRAPHY

- Steganography conceals secret information inside a normal-looking digital file.
- Images, audio, video, and text can be used as cover media.
- The resulting file containing hidden information is called a stego-file.
- At the receiving end, the hidden message is extracted using the agreed method or key.
- Hidden data may still be discovered through file inspection or statistical analysis.



STEGANOGRAPHY VS CRYPTOGRAPHY

- Cryptography changes the message into ciphertext; steganography hides the message in cover media.

❑ Cryptography protects the content, while steganography aims to hide that communication exists.

❑ Encrypted data may be recognizable as coded data; a stego-file is designed to look ordinary.

❑ Cryptanalysis studies encrypted messages; steganalysis looks for hidden information.

❑ Encrypting a message before hiding it can provide both confidentiality and concealment.



CONCLUSION

- Cryptography and steganography provide different but complementary forms of information security.
- Cryptography protects message content by making it unreadable without the key.
- Steganography conceals the presence of secret information in ordinary digital media.
- Both methods have limitations and must be implemented carefully.
- Used together, strong encryption and steganography can provide an additional layer of protection.



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